

EXPERIMENT 6

(A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

$\text{Difference} = \text{Apples Sold} - \text{Oranges Sold}$

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

CODE:

```
CREATE TABLE Sales
```

```
(
```

```
    sale_date DATE,
```

```
    fruit VARCHAR(20),
```

```
    sold_num INT,
```

```
    PRIMARY KEY (sale_date, fruit)
```

```
);
```

```
INSERT INTO Sales
```

```
VALUES
```

```
('2020-05-01','apples',10),
```

```
('2020-05-01','oranges',8),
```

```
('2020-05-02','apples',15),
```

```

('2020-05-02','oranges',15),
('2020-05-03','apples',20),
('2020-05-03','oranges',0),
('2020-05-04','apples',15),
('2020-05-04','oranges',16);
select sale_date,
sum(
    case
    when fruit='apples' then sold_num
    else -sold_num
    end
)AS diff
FROM Sales
GROUP BY sale_date
ORDER BY sale_date;
```

	sale_date date	diff bigint
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

(B) A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.

- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

CODE:

```
SELECT
  t.id,
CASE
  WHEN t.p_id IS NULL THEN 'Root'
  WHEN NOT EXISTS (
    SELECT 1
    FROM Tree c
    WHERE c.p_id = t.id
  ) THEN 'Leaf'
  ELSE 'Inner'
END AS Type
FROM Tree t
ORDER BY t.id;
```

	id [PK] integer	type text
1	1	Root
2	2	Inner
3	3	Leaf
4	4	Leaf
5	5	Leaf



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